The Architecture of Modern Immunology and Oncology: A Comprehensive Analysis of the Scientific Contributions and Life of Tak Wah Mak

The history of biological sciences in the late twentieth and early twenty-first centuries is marked by several "holy grails"—problems so complex they appeared insurmountable to the contemporary scientific community. In the realm of immunology, no mystery was more profound than the identity of the T-cell receptor (TCR). The resolution of this mystery, and the subsequent transformation of oncology through the lens of genetics, metabolism, and neurobiology, is inextricably linked to the life and work of Tak Wah Mak. A Canadian medical researcher, geneticist, and biochemist, Mak's career is defined by a series of intellectual pivots that have redefined the boundaries of human physiology.[1] His trajectory—from a childhood in Hong Kong to the pinnacle of global cancer research in Toronto—serves as a narrative of strategic non-conformity, persistent inquiry, and the rigorous application of molecular techniques to the "black boxes" of medicine.[3]

Formative Influences and the Non-Conventional Beginning

The intellectual foundations of Tak Wah Mak were established in an environment that prioritized discipline, classical education, and commercial pragmatism. Born on October 4, 1946, in Guangzhou, southern China, and raised in the British colony of Hong Kong, Mak was the son of silk merchants.[2] This familial background in the silk trade, centered in the Shunde district, was representative of a merchant class that dominated early Cantonese emigration and global trade networks.[6] While his parents encouraged him toward a career in medicine—a standard expectation for high-achieving youth in post-war Hong Kong—Mak's interests were rooted elsewhere, primarily in history, mathematics, biology, and chemistry.[2]

His early education was conducted at Wah Yan College in Kowloon, a highly regarded Jesuit school known for its competitive entrance and rigorous academic standards.[5] The Jesuit pedagogical tradition, emphasizing critical thinking and the synthesis of disparate ideas, likely played a role in shaping his eventual scientific philosophy. Upon graduating high school, Mak's primary goal was to avoid medical school, an ambition he fulfilled by pursuing chemical engineering and biochemistry in North America.[5]

The Transition to North America: Madison and Edmonton

The move to the United States in the mid-1960s coincided with a period of intense social and political upheaval. Mak's choice of university was influenced by a desire to avoid the anti-war activities that characterized the era. His mother and her associates discouraged him from attending the University of California, Berkeley, fearing the political climate would hinder his studies.[5] Consequently, he enrolled at the University of Wisconsin, Madison, which ironically possessed an anti-war movement even more radical than Berkeley's.[5] Despite the political environment, Wisconsin provided an elite academic foundation, housing one of the top chemistry and biochemistry departments in the United States.

Mak initially studied chemical engineering but soon recognized that his intellectual temperament was better suited to the foundational questions of biochemistry. He earned a Bachelor of Science in Chemistry/Biochemistry in 1967 and a Master of Science in Biophysics in 1969.[5] His move to Canada followed shortly thereafter, as he pursued doctoral studies at the University of Alberta in Edmonton. He completed his PhD in biochemistry in 1972, focusing on the biochemical properties of viral replication—a specialty that would serve as his entry point into the world of cancer research.[2]

Table 1: Primary Academic and Professional Timeline of Tak Wah Mak

Period	Institutional Affiliation	Degree or Role	Primary Field of Inquiry
1963–1967	University of Wisconsin, Madison	Bachelor of Science	Chemistry & Biochemistry [5]
1967–1969	University of Wisconsin, Madison	Master of Science	Biophysics [10]
1969–1972	University of Alberta, Edmonton	PhD	Biochemistry [5]
1972–1974	University of Toronto	Postdoctoral Fellow	Virology & Stem Cell Biology [2]
1984–Present	University of Toronto	Professor	Medical Biophysics & Immunology [2]
1993–2002	Amgen Research Institute	Director	Genetic Engineering & Knockout Models [11]

2004–Present	Campbell Family Institute	Director	Breast Cancer Research [2]

The Toronto Crucible: Mentorship and the Integration of Thought

Upon arriving in Toronto in the early 1970s, Mak joined the Ontario Cancer Institute (OCI), an environment that was then at the vanguard of cellular biology. His intellectual development was shaped by three primary mentors: James Till, Ernest McCulloch, and Howard Temin.[2] Till and McCulloch, the discoverers of hematopoietic stem cells, provided a masterclass in scientific rigor and originality. McCulloch was characterized as an intuitive thinker who could integrate complex thoughts into original ideas, whereas Till was a physicist who insisted on statistical significance and the absolute re-confirmation of all experimental results.[4] This duality—combining McCulloch's visionary integration with Till's rigorous verification—became the hallmark of Mak's own laboratory practice.[4]

Howard Temin, who visited Toronto and served as a mentor to Mak, provided another critical influence. Temin's discovery of reverse transcriptase had recently overturned the central dogma of molecular biology, demonstrating that information could flow from RNA to DNA. Temin encouraged Mak to look beyond the immediate constraints of any single discipline and to remain open to "diverse disciplines".[2] This interdisciplinary openness would prove vital when Mak transitioned from virology to immunology in the early 1980s.

The Search for the Holy Grail: Cloning the T-Cell Receptor

In the late 1970s and early 1980s, the central mystery of immunology was the identity of the T-cell receptor. While the B-cell receptor (the antibody) had been well-characterized, the mechanism by which T-cells recognized and destroyed infected or malignant cells remained unknown.[12] The search was so arduous and the failures so frequent that the TCR was referred to as the "Holy Grail of Immunology".[2] Early theories suggested that the TCR might be an unusual form of an antibody, but no one could isolate the gene or the protein.[10]

The Methodology of Molecular Subtraction

Mak's breakthrough was predicated on his background in virology, specifically the use of molecular subtraction techniques.[2] The premise was conceptually simple but technically demanding: because T-cells and B-cells were both lymphoid cells sharing the vast majority of their genome, the genes unique to T-cell function (such as the TCR) could be isolated by "subtracting" the shared genetic material.[2] By creating a library of T-cell-specific cDNA and removing sequences that hybridized with B-cell mRNA, Mak's team aimed to isolate the

"needle in the haystack."

The discovery was not without its obstacles. Mak famously worked against significant odds, including the rejection of a key research grant intended to fund the search.[5] Furthermore, the competitive landscape was intense, with laboratories led by Nobel laureates and established giants like Susumu Tonegawa and Ellis Reinherz also pursuing the receptor.[13] In 1984, the breakthrough finally arrived. A postdoctoral researcher in Mak's lab, Yusuke Yanagi, identified a unique cDNA clone (YT35) from human T-cells that encoded a protein with a molecular weight of 30 kilodaltons.[2]

The Race and Back-to-Back Publication

The discovery occurred in a near-simultaneous fashion with the work of Mark Davis and Stephen Hedrick at Stanford University.[2] While Mak's group identified the human TCR beta chain, Davis and Hedrick isolated the parallel gene in the mouse.[12] On March 8, 1984, Nature published their seminal findings in back-to-back articles.[2] This coincidence of discovery provided immediate cross-species validation, confirming that the identified genes were indeed the elusive receptors.

Mak's work revealed that T-cells use an entirely different set of genes for antigen recognition than B-cells, although the TCR was analogous in design to antibodies.[10] Shortly thereafter, Mak's lab demonstrated that the TCR genes underwent rearrangement in T-cell leukemia and lymphoma, providing a diagnostic tool for assessing the clonality and lineage of lymphoid malignancies.[12]

Table 2: Comparative Structural and Genetic Features of BCR and TCR

Feature	**B-Cell Receptor (BCR/Antibody)**	**T-Cell Receptor (TCR)**
Genetic Loci	Heavy (H), Light (L) Chains	Alpha (α), Beta (β), Gamma (γ), Delta (δ) Chains [12]
Recombination	V(D)J Gene Reassembly	V(D)J Gene Reassembly [16]
Secreted Form	Yes (Antibodies)	No (Strictly Membrane-bound) [17]
Antigen Recognition	Native structures (proteins,	Peptide fragments on MHC

	lipids)	molecules [12]
Discovery Date	Early-to-mid 20th Century	1984 (Mak and Davis) [2]

The impact of this discovery on the field of immunology cannot be overstated. It transformed T-cell biology from a series of observational experiments into a molecular science.[1] It also heralded the eventual development of Chimeric Antigen Receptor (CAR)-T and TCR-T cell technologies, which are now approved for the treatment of various leukemias and lymphomas.[1]

The Genetic Revolution: Pioneering the Knockout Mouse

Following the cloning of the TCR, Mak's focus shifted to functional genomics. He recognized that the ultimate proof of a gene's function lay in its behavior within a living system. This realization led his laboratory to become a global pioneer in the generation of "knockout mice"—genetically modified strains in which specific genes are inactivated to study the resulting phenotypes.[1]

Supported by a landmark partnership with the biotechnology firm Amgen, which led to the establishment of the Amgen Research Institute in Toronto in 1993, Mak's team generated more knockout mice than any other laboratory in the world.[2] This capability allowed them to systematically dissect the molecular pathways of the immune system and cancer. The Mak lab's work in this period was instrumental in elucidating the functions of several critical tumor suppressors and oncogenes.

Elucidating Tumor Suppressors: p53 and PTEN

The team used mutant animals to clarify the roles of p53 and PTEN, two of the most frequently mutated genes in human cancer.[1] They identified that Chk2 acts as a kinase that phosphorylates and stabilizes p53, allowing it to execute its anti-tumorigenic functions, such as apoptosis or cell cycle arrest in response to DNA damage.[22]

In the case of PTEN, the Mak lab demonstrated that this molecule negatively regulates the PKB/AKT cell survival pathway.[22] PTEN-deficient cells were found to be resistant to apoptosis-inducing agents, effectively becoming immortalized and susceptible to further tumorigenic mutations.[22] More recent work has shown that PTEN and ATM (Ataxia-Telangiectasia Mutated) cooperate to regulate the fitness and quiescence of hematopoietic stem cells, and that this same PTEN-ATM axis controls cell cycle checkpoints in HER2+ breast cancer.[23]

Dissecting Breast Cancer Genes: BRCA1 and BRCA2

The lab also made significant contributions to the understanding of BRCA1 and BRCA2, genes that are central to hereditary breast and ovarian cancer. By generating mouse models with specific mutations in these genes, Mak's team helped define how their loss leads to genomic instability and tumorigenesis, providing a foundation for the development of targeted therapies like PARP inhibitors.[1]

The Discovery of the Immune "Brake": CTLA-4 and the Birth of Checkpoint Inhibitors

Perhaps the most clinically significant discovery resulting from the Mak lab's knockout mouse program was the identification of the function of CTLA-4 (Cytotoxic T-Lymphocyte-Associated protein 4). In 1995, Mak's group published a landmark paper showing that CTLA-4 acts as a negative regulator of T-cell activation.[1]

The Mechanism of CTLA-4 Signaling

The T-cell receptor provides "Signal 1" for activation, but T-cells also require a co-stimulatory "Signal 2," typically provided by the interaction of CD28 on the T-cell with B7 ligands on an antigen-presenting cell.[13] CTLA-4 was discovered to compete with CD28 for these same B7 ligands, but with much higher affinity.[13] When CTLA-4 binds B7, it sends an inhibitory signal that halts T-cell proliferation and effector function.

In Mak's knockout mice, the absence of this "brake" led to a catastrophic autoimmune phenotype, where T-cells became hyper-activated and infiltrated every organ of the body, leading to early death.[1] This provided "absolute genetic proof" that CTLA-4 was a critical checkpoint for maintaining immune self-tolerance.[25]

From Bench to Bedside: Ipilimumab

While Mak focused on the fundamental biology, James Allison at the University of Texas M.D. Anderson Cancer Center recognized the potential to use this "brake" to treat cancer.[13] Allison hypothesized that by blocking the CTLA-4 molecule with an antibody, one could "take the brakes off" the immune system and allow it to attack tumors that would otherwise evade detection.[14] This concept of immune checkpoint blockade led to the development of Ipilimumab, the first in a new class of immunotherapies that have since revolutionized the treatment of melanoma and many other cancers.[14]

A Strategic Pivot to Cancer Metabolism: IDH1 and IDH2

In the early 2000s, Mak once again demonstrated his willingness to challenge conventional research paths. Despite his status as a world leader in immunology, he sensed that the field was reaching a point of diminishing returns in terms of novel drug targets. He turned his attention to cancer metabolism, a field that had been relatively dormant since the discovery of the Warburg Effect in the early 20th century.[2]

This pivot was formalized in 2008 with the co-founding of Agios Pharmaceuticals, a biotech company dedicated to targeting metabolic alterations in cancer.[2] Working with collaborators like Craig Thompson and Lewis Cantley, Mak sought to identify "oncometabolites"—small molecules produced by mutated enzymes that drive cancer progression.

The Neomorphic Activity of Mutant IDH

The research focused on Isocitrate Dehydrogenase 1 and 2 (IDH1 and IDH2), enzymes that normally catalyze the conversion of isocitrate to α-ketoglutarate (α-KG) while producing NADPH.[28] In 2009, a separate cancer genome project identified mutations in IDH1 in patients with glioblastoma and acute myeloid leukemia (AML).[28]

Mak's team and the scientists at Agios discovered that these mutations—specifically at arginine 132 in IDH1 and equivalent residues in IDH2—were not simple "loss-of-function" mutations.[28] Instead, they conferred a "neomorphic" gain-of-function activity: the mutant enzymes began to reduce α-KG into a rare metabolite called D-2-hydroxyglutarate (D-2-HG).[29]

D-2-Hydroxyglutarate as an Oncometabolite

The accumulation of D-2-HG was found to be highly toxic to the cell's epigenetic regulatory machinery. D-2-HG acts as a competitive inhibitor of multiple α-KG-dependent dioxygenases, including the TET family of DNA methylcytosine dioxygenases and histone demethylases.[30] This leads to a global state of hypermethylation, blocking the differentiation of hematopoietic stem cells and promoting leukemogenesis.[32]

Table 3: Summary of IDH1/2 Mutation Consequences

Factor	Normal IDH1/2 Function	Mutant IDH1/2 Function (Neomorphic)
Substrate	Isocitrate	α-ketoglutarate (α-KG) [29]

Main Product	α-ketoglutarate (α-KG)	D-2-hydroxyglutarate (D-2-HG) [29]
Cofactor Role	Produces NADPH	Consumes NADPH [29]
Downstream Effect	Supports cellular redox and lipid synthesis	Inhibits TET2 and histone demethylases [32]
Clinical Outcome	Homeostasis	Blocked differentiation and tumorigenesis [28]

This discovery led to the development of Enasidenib (IDHIFA), a first-in-class drug that specifically inhibits the mutant IDH enzyme. Approved by the FDA in 2017 for relapsed or refractory AML, IDHIFA became the first licensed drug to specifically target cancer metabolism, providing a new pillar of therapy for patients who had previously failed standard treatments.[2]

The Neuro-Immune Synthesis: The Cholinergic Revolution

In his most recent phase of investigation, Mak has explored the interconnection between the nervous system and the immune system. For more than a century, scientists have pondered how the brain communicates with the body's defenses, but the molecular bridges remained hidden. In 2019, the Mak lab published a series of findings in Science that solved a critical piece of this puzzle.[25]

Acetylcholine-Producing T-Cells

The researchers demonstrated that a subset of T-cells and B-cells can synthesize acetylcholine, a molecule classically known as a neurotransmitter that controls learning, memory, and muscle activation in the brain.[1] They identified that these "cholinergic T-cells" express the enzyme choline acetyltransferase (ChAT), which allows them to produce acetylcholine in response to neural signals or chronic infection.[25]

Mechanism of Action: The Vasodilation Relay

The production of acetylcholine by T-cells was found to be essential for antiviral immunity. During a viral infection, T-cells must migrate from the bloodstream into the infected tissues. Mak's team showed that T-cell-derived acetylcholine triggers local vasodilation—the widening of blood vessels—which slows down the blood flow and allows the T-cells to "jump out" of the circulation and reach the site of infection.[25]

Using genetically engineered mice that lacked the ability to produce acetylcholine in their T-cells, the Mak lab observed that these animals could not control chronic viral infections.[25] Without the neurotransmitter, the immune cells were effectively trapped in the blood vessels, unable to infiltrate the tissues to kill virus-infected cells.[40]

Implications for Autoimmunity and Cancer

This discovery has broad implications for diverse diseases. In cancer, tumors often surround themselves with defenses that prevent immune cell entry; Mak suggests that strategies to increase immune neurotransmitter production could help T-cells penetrate the tumor microenvironment.[25] Conversely, in autoimmune diseases like rheumatoid arthritis or multiple sclerosis, where T-cells inappropriately attack self-tissues, a reduction in acetylcholine signaling might help "quell the hordes" of immune cells invading the joints or the central nervous system.[25]

Table 4: The Role of Acetylcholine in the Neuro-Immune Axis

System	Signal Origin	Target Receptor/Effect	Outcome
Brain	Cholinergic Neurons	Nicotinic/Muscarinic Receptors	Learning, Memory, Arousal [36]
Immune (Antiviral)	ChAT+ T-cells	Endothelial Receptors (Vasodilation)	Tissue Infiltration by T-cells [25]
Inflammatory Reflex	Vagus Nerve/Splenic T-cells	α7nAChR on Macrophages	Suppression of TNF-α and inflammation [39]
Cancer	T-cell Synapses	Tumor Microenvironment	Improved immune surveillance (potential) [38]

Scientific Philosophy and Mentorship: The "Seed" Theory

Tak Wah Mak's impact is measured not only by his personal discoveries but also by his role as a mentor. Over the course of nearly 40 years at the University of Toronto and the Princess

Margaret Cancer Centre, he has trained more than 150 students and postdoctoral fellows.[3] His approach to mentorship is rooted in the belief that the future of science depends on the "seeds" planted today.

The Two Lives of a Medical Scientist

Mak often cites a modified version of a quote attributed to Confucius: "We all have two lives, and the second one begins when we realize we only have one".[3] In Mak's version, "All medical scientists have two lives, and the second one begins when they realize that immunology is the orchestra for most of life's symphonies".[3] This philosophy emphasizes the centrality of the immune system in understanding not just cancer, but neurodegeneration, cardiac disease, and aging.

The "Sinek Principle" and the Question of "Why"

Drawing on the ideas of author Simon Sinek, Mak encourages his trainees to "always start with Why".[3] He argues that while many researchers focus on *what* to do or *how* to do it, the most significant breakthroughs come from asking *why* certain phenomena occur—such as why some patients respond dramatically to immunotherapy while others do not.[3] This persistent digging into "black boxes" is what Mak believes propels science forward.

Commitment to Canadian Science: A National Identity

One of the most pivotal decisions in Mak's life occurred in 1987. Following the global acclaim for the TCR discovery, he was offered prestigious positions as Head of Pathology at Yale University and Director of the Howard Hughes Medical Institute.[2] He turned down both offers to remain in Toronto, a decision that led to him being described as a "national hero for not leaving".[43] Mak's reasoning was pragmatic: he believed that Canada offered a less hyper-competitive, "contact sport" environment than the United States, allowing for more innovative and integrated thinking.[43]

Recognition and Global Stature

The magnitude of Mak's contributions has been recognized by nearly every major international scientific body. He has received numerous prestigious awards, often for breakthroughs that fundamentally altered the course of their respective fields.

Table 5: Selected International Awards and Honors

Year	Award/Honor	Reasoning/Discovery Linked

1986	Fellow of the Royal Society of Canada	TCR Discovery and Genetic Immunology [8]
1989	Gairdner Foundation International Award	Breakthrough in T-cell Biology [2]
1994	Fellow of the Royal Society (London)	Fundamental contributions to Immunology [8]
1995	King Faisal International Prize for Medicine	Discovery of the TCR and Checkpoint Biology [2]
2000	Officer of the Order of Canada	Outstanding service to Canadian science [2]
2004	Paul Ehrlich and Ludwig Darmstaedter Prize	Pioneering work in T-cell recognition [2]
2009	Canadian Medical Hall of Fame	Lifelong contributions to medical research [2]
2021	Szent-Györgyi Prize	Progress in Cancer Research and Metabolism [2]
2023	Pezcoller Foundation-AACR Award	Extraordinary Achievement in Cancer Research [10]

These awards represent a recognition of Mak's unique ability to span the distance from basic molecular cloning to clinical drug development. His work has transitioned from identifying the "antenna" of the immune system (TCR) to the "brakes" (CTLA-4), and finally to the "engine" (metabolism) and the "wiring" (neuroimmunology).[3]

Pivotal Anecdotes: The Odds and the "Leap of Faith"

The narrative of Tak Mak's career is punctuated by moments of strategic intuition and defiance of consensus. One of the most telling anecdotes involves the 1984 TCR discovery. While the scientific community believed the receptor would be an antibody, Mak's intuition suggested otherwise. He famously made a sketch of what he believed the shape of the alpha and beta chains would look like for a student, Pamela Ohashi.[44] Ohashi kept the sketch, and years later, the structural predictions in that drawing were proven correct: the chains were indeed held together by intermolecular disulfide bonding.[18] Mak described this as a "leap of

faith" required when the facts are not yet fully in.[44]

Another pivotal moment was the establishment of the Amgen Research Institute. At a time when university funding was becoming more difficult, Mak negotiated a partnership that gave him $100 million in funding while maintaining academic independence.[10] This "private-academic" hybrid model allowed him to scale his knockout mouse program to a level that was previously unimaginable, leading to the identification of 20 patented molecular discoveries for use in drug development.[10]

The Future of Anti-Cancer Therapy: An Ongoing Mission

At age 79, Tak Mak remains an active and influential figure in science, with no plans to retire. He has stated, "I don't have anything left to prove - except that cancer is not cured yet".[10] His current work focuses on identifying novel cell-surface molecules that T-cells can recognize as cancer, aiming to expand the capabilities of CAR-T and TCR-T therapies.[26]

He also continues to investigate the cGAS/STING pathway—a mechanism of the innate immune system that detects cytosolic DNA as a sign of danger—and how it responds to the genotoxic stress caused by chemotherapy and radiation.[22] By understanding how these pathways cooperate with tumor suppressors like PTEN and p53, Mak aims to develop more precise, combined interventions that can overcome the resistance mechanisms of advanced cancers.[23]

Conclusion: The Integrated Legacy of Tak Wah Mak

The career of Tak Wah Mak provides a masterclass in the evolution of biomedical science from the cellular to the systemic. His life's work is characterized by a refusal to be confined by the traditional silos of academic research. By identifying the T-cell receptor, he unlocked the door to modern immunology; by pioneering knockout mice, he provided the tools for functional genomics; by discovering the role of CTLA-4, he enabled the revolution of immunotherapy; and by mapping the metabolic and neurological regulation of immune cells, he has opened new frontiers in the holistic treatment of disease.

Mak's legacy is defined by his ability to connect the dots between seemingly unrelated fields. Whether it is the fermentation of sugar in a cancer cell or the production of brain chemicals in a T-cell, Mak has consistently found the threads that unite these phenomena under a single biological framework. As he continues to mentor the next generation of "explorers," his commitment to asking "Why?" remains the driving force behind a career that has changed the face of oncology and provided hope to thousands of patients worldwide.[3] Through scientific excellence and an unwavering commitment to the "orchestra of life," Tak Wah Mak has ensured that the "flowers of tomorrow" will be rooted in the profound "seeds" of his

discoveries.[3]

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44. The Holy Grail of Immunology | The Scientist, accessed January 25, 2026, https://www.the-scientist.com/the-holy-grail-of-immunology-51335

SCENE #1: The Leap of Faith Summary: In a high-stakes effort to solve the "Holy Grail of Immunology," the laboratory uses an unconventional virology technique to clone the human T-cell receptor (TCR) beta chain, succeeding despite significant institutional barriers. Date/Place: March 8, 1984 (Publication); University of Toronto / Ontario Cancer Institute. People: Tak Wah Mak, Yusuke Yanagi (postdoctoral researcher), and Pamela Ohashi (student). Concrete details (quote-level if available): Facing a "rejected research grant" and working "against all odds," the team employed "molecular subtraction" to isolate T-cell-specific cDNA. Before the structure was fully confirmed, Mak took a "leap of faith" and hand-drew a sketch for Pamela Ohashi showing how he believed the α and β chains were held together by intermolecular disulfide bonding. Why it matters: It transformed immunology from observational biology into a molecular science and laid the groundwork for all modern T-cell therapies, including CAR-T and TCR-T. Evidence quality (primary/secondary; strength): Primary; Based on seminal Nature publications (1984) and personal accounts from lab members. Myth risk (low/medium/high + why): Medium; While the discovery was simultaneous with Mark Davis, the narrative of an "intense race" is sometimes heightened for drama, though they published back-to-back in the same issue.

SCENE #2: The Exploding Lymph Nodes Summary: The laboratory pioneers the use of "knockout mice" to determine the function of CTLA-4, discovering that rather than an activator, the molecule serves as a critical "brake" that prevents the immune system from destroying the body. Date/Place: 1995; Amgen Research Institute, Toronto. People: Tak Wah Mak, Paul Waterhouse (lab member), and Arlene Sharpe (independent collaborator/competitor). Concrete details (quote-level if available): The breakthrough came when researchers observed the physical phenotype of mice lacking the CTLA-4 gene. The animals became hyper-inflamed and "turned into a big lymph node" because their T-cells "couldn't stop dividing." This provided "absolute genetic proof" of the molecule's inhibitory function. Why it matters: This discovery identified the first immune "checkpoint," providing the biological rationale for checkpoint inhibitor drugs like Ipilimumab, which revolutionized cancer treatment. Evidence quality (primary/secondary; strength): Primary/Secondary; Documented in Science (1995) and confirmed in later Nobel Prize-related retrospectives. Myth risk (low/medium/high + why): Low; The dramatic phenotype of the knockout mice is a well-documented turning point in functional genomics.

SCENE #3: The National Hero’s Choice Summary: Following the international acclaim of the TCR discovery, Mak faces a pivotal career crossroad, ultimately choosing to remain in Canada despite lucrative offers to lead top-tier American institutions. Date/Place: 1987; Toronto / USA. People: Tak Wah Mak. Concrete details (quote-level if available): Mak was offered the positions of Head of Pathology at Yale University and Director of the Howard Hughes Medical Institute. He turned them down, leading to headlines calling him a "national hero for not leaving." He later remarked that science in the United States felt like a "contact sport" and preferred the innovative, less hyper-competitive environment of Toronto. Why it matters: This

decision solidified Toronto as a global center for immunology and genetics, ensuring that his laboratory's "seeds"—over 150 trainees—would continue to develop within the Canadian research ecosystem. Evidence quality (primary/secondary; strength): Secondary; Based on biographical profiles and interviews reflecting on his career trajectory and philosophy. Myth risk (low/medium/high + why): Low; The job offers and Mak's subsequent "national hero" label are widely cited in his institutional biographies.

Sources:

Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*. https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf , Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. https://www.cdnmedhall.ca/laureates/takmak , Tak Wah Mak. (2024). In *Wikipedia*. https://en.wikipedia.org/wiki/Tak_Wah_Mak , The Scientist. (2003, July 13). *The Holy Grail of Immunology*. https://www.the-scientist.com/the-holy-grail-of-immunology-51335 , Allison, J. P. (2018). *Nobel Lecture: Immune Checkpoint Blockade in Cancer Therapy*. https://www.nobelprize.org/uploads/2018/10/allison-lecture.pdf , Oldfield, J. (2019, May 23). *'Discover everything there is': U of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research*. University of Toronto News. https://www.utoronto.ca/news/discover-everything-there-u-t-s-tak-mak-awarded-prestigious-gold-leaf-prize-pioneering-research , Government of Canada. (2002). *Tak W. Mak propels his University of Toronto lab to the forefront of cancer research*. Publications.gc.ca. https://publications.gc.ca/collections/collection_2022/isde-ised/iu4/Iu4-8-2002-eng.pdf , National Academy of Sciences. (n.d.). *Profile of Tak Wah Mak*. Proceedings of the National Academy of Sciences. https://www.pnas.org/doi/10.1073/pnas.1116912108 , Cancer World. (n.d.). *The future of cancer treatment: Tak Mak, T-cell receptor, immunotherapies*. https://cancerworld.net/the-future-of-cancer-treatment-tak-mak-t-cell-receptor-immunotherapies/ , OncoDaily. (2024). *40th Anniversary of the Discovery of the T Cell Receptor*. https://oncodaily.com/insight/tcr179986 , University Health Network. (2023, March 14). *Honouring Extraordinary Achievement*. https://www.uhnresearch.ca/news/honouring-extraordinary-achievement

1. On the Centrality of Immunology

- **Exact Quote:** "All medical scientists have two lives. The second one begins when they realize that immunology is the orchestra for most of life's symphonies."
- **Context/Date:** 2023 interview with the Canadian Cancer Research Alliance (CCRA).
- **Where it appears:** CCRA Interview Transcript, *2023 Interview with Dr. Tak Wah Mak*.
- **Full Citation:** Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak*

Wah Mak. https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf

2. On Scientific Intuition (The "Leap of Faith")

- **Exact Quote:** "At the very beginning, when thoughts are still brewing in your mind, and the facts are not yet in, you have to take a leap of faith. And luckily, we turned out to be right."
- **Context/Date:** July 13, 2003; reflecting on his 1984 decision to sketch the predicted structure of the T-cell receptor before its final confirmation.
- **Where it appears:** *The Scientist* magazine, article titled "The Holy Grail of Immunology".
- **Full Citation:** The Scientist. (2003, July 13). *The Holy Grail of Immunology*. https://www.the-scientist.com/the-holy-grail-of-immunology-51335

3. On the Joy of Discovery

- **Exact Quote:** "When you're a kid you get to play with toys and figure things out, and you're not supposed to do that as an adult. But science is like toys for adults, and we get to discover everything there is."
- **Context/Date:** May 23, 2019; interview following the announcement of the CIHR Gold Leaf Prize for Discovery.
- **Where it appears:** University of Toronto News article, *'Discover everything there is': U of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research*.
- **Full Citation:** Oldfield, J. (2019, May 23). *'Discover everything there is': U of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research*. University of Toronto News. https://www.utoronto.ca/news/discover-everything-there-u-t-s-tak-mak-awarded-prestigious-gold-leaf-prize-pioneering-research

4. On Retirement and Persistence

- **Exact Quote:** "I don't have anything left to prove - except that cancer is not cured yet."
- **Context/Date:** March 2023; statement regarding his continued research at age 76 after receiving the Pezcoller Foundation-AACR Award.
- **Where it appears:** Canadian Medical Hall of Fame Laureate Biography.
- **Full Citation:** Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. https://www.cdnmedhall.ca/laureates/takmak

5. On His Trainees (The "Seed Theory")

- **Exact Quote:** "The flowers of all the tomorrows are in the seeds of today. I like to think of my trainees as those seeds, who will set up their own labs and continue to spread flowers in the form of useful research discoveries."
- **Context/Date:** 2023; interview regarding his legacy and the training of over 150 postdoctoral fellows and students.
- **Where it appears:** CCRA Interview Transcript, *2023 Interview with Dr. Tak Wah Mak*.
- **Full Citation:** Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*. https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf

6. On Remaining in Canada

- **Exact Quote:** "Only in Canada does someone become a national hero for not leaving. Toronto is a good place to do science. It's less of a contact sport here than it is in the United States."
- **Context/Date:** 2002; commenting on his decision in 1987 to turn down prestigious leadership roles at Yale and the Howard Hughes Medical Institute.
- **Where it appears:** Government of Canada Publication, *Tak W. Mak Propels His University of Toronto Lab to the Forefront of Cancer Research*.
- **Full Citation:** Government of Canada. (2002). *Tak W. Mak propels his University of Toronto lab to the forefront of cancer research*. Publications.gc.ca. https://publications.gc.ca/collections/collection_2022/isde-ised/iu4/Iu4-8-2002-eng.pdf

7. On Mentorship and Rigor

- **Exact Quote:** "McCulloch taught me to think differently, to have original ideas and integrate my thoughts. Till was a staunch, vigorous physicist, who insisted that all results be statistically significant and re-confirmed. The two together made an almost perfect mentorship for me."
- **Context/Date:** February 2019; reflecting on his early years at the Ontario Cancer Institute working with James Till and Ernest McCulloch.
- **Where it appears:** University of Toronto Faculty of Medicine News, *Tak Mak: Gold Leaf Prize and Life Science*.
- **Full Citation:** University of Toronto. (2019, February). *Tak Mak: Gold Leaf Prize and Life Science*. https://temertymedicine.utoronto.ca/news/tak-mak-gold-leaf-prize-and-life-science

As a scientist, Tak Wah Mak is characterized by a blend of Jesuit-influenced rigor, merchant-class pragmatism, and a deliberate avoidance of "scientific crowds."

Human Reality Details

- **Upbringing/Constraint:** Born into a family of silk merchants in post-war Hong Kong, Mak faced a culture where high-achieving youth were strictly expected to become doctors, lawyers, or businessmen; he chose biochemistry primarily to "avoid medical school".
- **Formative Experience/Jesuit Education:** His education by Jesuit priests at Wah Yan College instilled a "creed" of benevolence and fairness that shaped his eventual scientific principles of inquiry and persistence.
- **Obstacle/Early Struggle:** When he first arrived in the United States, his family had very little money, forcing him to cover his tuition by working as a gardener and a glassware washer in research facilities.
- **Working Style/Non-Conformity:** As a student, Mak displayed a pragmatic defiance of academic busywork, once solving only three out of 100 assigned math problems and arguing that solving the remaining 97 was unnecessary if his initial logic was correct.
- **Mentor Influence:** Mak's scientific rigor was forged by a "perfect" dual mentorship: Ernest McCulloch taught him to think intuitively and "big picture," while James Till, a physicist, insisted on absolute re-confirmation of every result.
- **Working Habit/Retreats:** He fostered a collaborative culture by hosting lab "leukaemia meetings" at family cottage retreats on Ahmic Lake, combining sailing and outdoor activities with intense biological discussion.
- **Scientific Philosophy/Solitude:** Mak intentionally avoids "scientific crowds," believing that a researcher must be "lonely" enough to retreat from the group and evaluate ideas with their own unique bias to find new paths.
- **Mentoring Style/Legacy:** He views his trainees as "seeds" rather than just employees, taking more pride in the 150+ researchers he has mentored into leadership roles than in his own individual awards.
- **Working Style/Intuition:** He is a proponent of taking a "leap of faith" in research, famously hand-sketching the T-cell receptor's alpha-beta chain structure for a student years before the physical evidence was finalized.
- **Cultural Identity/National Decision:** Mak rejected leadership offers from Yale and the Howard Hughes Medical Institute because he preferred Canada's less hyper-competitive environment, which he described as less of a "contact sport" than U.S. science.
- **Quirks/Proverbs:** He frequently frames complex scientific strategies using proverbs, such as the African proverb "If you want to go far, go together," to emphasize teamwork in tackling cancer.
- **Habit/Persistence:** At 79 years old, he continues to lead active research projects, famously stating he has nothing left to prove except that "cancer is not cured yet".

Tak Wah Mak's research provides a direct bridge to introductory biochemistry, specifically within the study of **Metabolism and Enzyme Catalysis**.

BIOL 3030 Bridge

- **Best-Fitting Topic:** Metabolism and Enzyme Catalysis (specifically the Citric Acid Cycle and Redox Cofactors).
- **Precise Contribution:** Dr. Mak discovered that mutations in the metabolic enzymes Isocitrate Dehydrogenase 1 and 2 (IDH1/2) confer a "neomorphic" gain-of-function activity, producing the oncometabolite D-2-hydroxyglutarate (D-2-HG) which drives cancer progression.
- **Teachable Bridge:** In introductory biochemistry, students learn that IDH is a key enzyme in cellular respiration that catalyzes the oxidative decarboxylation of isocitrate to α-ketoglutarate (α-KG) while producing $NADPH$. Mak's work demonstrates how a single amino acid substitution in the enzyme's active site causes it to consume $NADPH$ to reduce α-KG into D-2-HG. This rare metabolite then acts as a competitive inhibitor for other α-KG-dependent enzymes, such as DNA demethylases, illustrating how a specific change in enzyme kinetics and substrate specificity can rewire a cell's entire epigenetic landscape to promote malignancy.

Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*.(https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf) , Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. https://www.cdnmedhall.ca/laureates/takmak , El País. (2024, September 22). *Tak Mak, immunologist: 'The immune system is the orchestra for most of life's symphonies'*. https://english.elpais.com/health/2024-09-22/tak-mak-immunologist-the-immune-system-is-the-orchestra-for-most-of-lifes-symphonies.html , Frontline Genomics. (2024). *Genome Giants: Tak Mak, Senior Scientist, Princess Margaret Cancer Centre*. https://frontlinegenomics.com/genome-giants-tak-mak-senior-scientist-princess-margaret-cancer-centre/ , Government of Canada. (2002). *Tak W. Mak propels his University of Toronto lab to the forefront of cancer research*. Publications.gc.ca. https://publications.gc.ca/collections/collection_2022/isde-ised/iu4/Iu4-8-2002-eng.pdf , Mak, T. W. (2018). How I remember McCulloch. *Biographical Memoirs of Fellows of the Royal Society, 64*, 317–324. https://doi.org/10.1098/rsbm.2017.0033 , National Academy of Sciences. (2011). *Profile of Tak Wah Mak*. *Proceedings of the National Academy of Sciences, 108*(2), 11169-11171. https://doi.org/10.1073/pnas.1116912108 , Oldfield, J. (2019, May 23). *'Discover everything there is': U of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research*. University of Toronto News. https://www.utoronto.ca/news/discover-everything-there-u-t-s-tak-mak-awarded-prestigious-gold-leaf-prize-pioneering-research , Spanish National Cancer Research Center (CNIO). (2023, March 21). *Tak W. Mak, immunologist: "I am not doing science for recognition"*. https://www.cnio.es/en/news/tak-w-mak-immunologist-i-am-not-doing-science-for-recognition/ , The Scientist. (2003, July 13). *The Holy Grail of Immunology*.

https://www.the-scientist.com/the-holy-grail-of-immunology-51335 , University of Toronto. (2019, February). *Tak Mak: Gold Leaf Prize and Life Science*. https://temertymedicine.utoronto.ca/news/tak-mak-gold-leaf-prize-and-life-science , Wikipedia. (2024). *Tak Wah Mak*.(https://en.wikipedia.org/wiki/Tak_Wah_Mak) .

Scientist's Analogy

When describing the metabolic mutations in cancer cells that allow them to adapt to their environment and evade death, Mak characterizes this strategic biological shift as "**shaking hands with the devil**." He also frequently employs a mechanical analogy for the immune system, where the T-cell receptor is the **ignition**, the CD28 molecule is the **accelerator**, and the CTLA-4 molecule (which he co-discovered) is the **brake**.

Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*. https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf , Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. https://www.cdnmedhall.ca/laureates/takmak , Wikipedia. (2024). *Tak Wah Mak*. https://en.wikipedia.org/wiki/Tak_Wah_Mak , The Scientist. (2003, July 13). *The Holy Grail of Immunology*. https://www.the-scientist.com/the-holy-grail-of-immunology-51335 , Allison, J. P. (2018). *Nobel Lecture: Immune Checkpoint Blockade in Cancer Therapy*. https://www.nobelprize.org/uploads/2018/10/allison-lecture.pdf , Oldfield, J. (2019, May 23). *'Discover everything there is': U of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research*. University of Toronto News. https://www.utoronto.ca/news/discover-everything-there-u-t-s-tak-mak-awarded-prestigious-gold-leaf-prize-pioneering-research , Government of Canada. (2002). *Tak W. Mak propels his University of Toronto lab to the forefront of cancer research*. Publications.gc.ca. https://publications.gc.ca/collections/collection_2022/isde-ised/iu4/Iu4-8-2002-eng.pdf , National Academy of Sciences. (n.d.). *Profile of Tak Wah Mak*. Proceedings of the National Academy of Sciences. https://www.pnas.org/doi/10.1073/pnas.1116912108 , Cancer World. (n.d.). *The future of cancer treatment: Tak Mak, T-cell receptor, immunotherapies*. https://cancerworld.net/the-future-of-cancer-treatment-tak-mak-t-cell-receptor-immunotherapies/ , OncoDaily. (2024). *40th Anniversary of the Discovery of the T Cell Receptor*. https://oncodaily.com/insight/tcr179986 , University Health Network. (2023, March 14). *Honouring Extraordinary Achievement*. https://www.uhnresearch.ca/news/honouring-extraordinary-achievement , The Princess Margaret Cancer Foundation. (2024, November 21). *Celebrating a revolution in immunology and cancer care*. https://thepmcf.ca/media-supporter-stories/latest-news/2024/celebrating-a-revolution-in-immunology-and-cancer-care/ , Chinese Australia. (2024). *Shunde silk production*. https://chineseaustralia.org/2024/01/.

Tak Wah Mak's career has been marked by several significant priority disputes, institutional

barriers, and documented investigations that reflect the high-stakes nature of modern immunology and oncology.

1. The T-Cell Receptor Priority Race and Funding Rejection

- **What's agreed upon:** In the early 1980s, the race to clone the T-cell receptor (TCR) was considered the "Holy Grail of Immunology." Most established groups, including those led by Ellis Reinherz and James Allison, attempted to isolate the receptor via protein purification. Tak Mak and Mark Davis ultimately won the race by using the "molecular subtraction" technique at the RNA/DNA level.
- **What's disputed:** Mak faced significant institutional skepticism regarding his approach. His initial grant proposal to fund the search was flatly rejected, as reviewers considered the proposed molecular subtraction method "impossible" or unlikely to succeed. Furthermore, while Mak and Davis are credited with the genetic discovery, other scientists like Reinherz had previously identified the protein's biochemical components, leading to long-standing discussions regarding who provided the "true" discovery.
- **Best sources:** *Nature* (1984) for the discovery; *PNAS* for the rejected grant narrative.

2. The CTLA-4 Function Controversy and the 2018 Nobel "Snub"

- **What's agreed upon:** James Allison and Tasuku Honjo received the 2018 Nobel Prize for discovering cancer therapy via immune checkpoint blockade. Mak's lab, alongside Arlene Sharpe's, provided the "absolute genetic proof" in 1995 that CTLA-4 acts as an inhibitory "brake" on T-cells.
- **What's disputed:** A priority dispute existed in the mid-1990s regarding whether CTLA-4 was an activator or an inhibitor. Earlier researchers, such as Peter Linsley, initially suggested CTLA-4 was a co-stimulatory "gas pedal" similar to CD28. Mak and Sharpe independently overturned this "heretical" view. When the 2018 Nobel was awarded to Allison (who applied the brake theory to cancer), many experts were "shocked" that the genetic discoverers—specifically Mak, Sharpe, and Gordon Freeman—were excluded due to the Nobel "Rule of 3."
- **Best sources:** *Science* (1995) for the knockout data; *STAT News* and *New York Times* for the 2018 Nobel exclusion controversy.

3. Investigation of Alleged Scientific Misconduct

- **What's agreed upon:** Tak Mak and his high-profile protégé, Josef Penninger, were the subjects of a documented institutional investigation.
- **What's disputed:** The investigation focused on alleged scientific misconduct. While Penninger eventually moved to lead the Institute of Molecular Biotechnology in Austria, the publicly documented fallout was described by peers as a significant "falling-out" between elite scientists that captured the attention of the Canadian research

community.
- **Best sources:** *Immpress Magazine*; *Schneider Shorts* (2025 Retrospective).

4. Agios Intellectual Property Lawsuit

- **What's agreed upon:** Tak Mak co-founded Agios Pharmaceuticals with Craig Thompson and Lewis Cantley to target cancer metabolism. In 2011-2012, the University of Pennsylvania and the Abramson Institute filed lawsuits against Thompson and the company.
- **What's disputed:** The lawsuit alleged that Thompson had "absconded" with intellectual property developed at Penn related to IDH1 and IDH2 mutations to found Agios. While Thompson was the primary target of the "unscrupulous" label in the filings, the dispute threatened the reputation and financial foundations of the company Mak helped build. The defense argued that authorship of papers did not equate to patentable "invention" under the specific circumstances.
- **Best sources:** *The Cancer Letter* archive.

Here are the APA-formatted sources for the complications, disputes, and controversies involving Tak Wah Mak:

1. The T-Cell Receptor Priority Race and Funding Rejection

- **Grant Rejection:** National Academy of Sciences. (2011). Profile of Tak Wah Mak. *Proceedings of the National Academy of Sciences, 108*(27), 11169-11171.
- **Discovery Priority:** Yanagi, Y., Yoshikai, Y., Leggett, K., Clark, S. P., Aleksander, I., & Mak, T. W. (1984). A human T cell-specific cDNA clone encodes a protein having extensive homology to immunoglobulin chains. *Nature, 308*(5955), 145-149.
- **Historical Competition:** Odaka, C. (2020). PD-1: Its discovery, involvement in cancer immunotherapy, and current research. *Cells, 9*(6), 1376.

2. The CTLA-4 Function Controversy and the 2018 Nobel "Snub"

- **Genetic Proof:** Waterhouse, P., Penninger, J. M., Holliday, E., Ohashi, P. S., & Mak, T. W. (1995). Lymphoproliferative disorders with early lethality in mice deficient in CTLA-4. *Science, 270*(5238), 985-988.
- **Nobel Exclusion Controversy:** Advisory Board. (2018, October 4). *Were 3 scientists 'snubbed' by the Nobel Prize in medicine?*
- **Scientific Background of Snub:** Bluestone, J. A. (2016). Checkpoint blockade: The triumph of basics. *Science Signaling, 9*(415).

3. Investigation of Alleged Scientific Misconduct

- **Investigation Details:** Immpress Magazine. (2024, June 3). *Ego, politics, and the*

scientific zeitgeist.

- **Legacy Context:** Schneider, L. (2025, November 7). Schneider Shorts 7.11.2025: No data were falsified or fabricated in any way. *For Better Science*.

4. Agios Intellectual Property Lawsuit

- **Lawsuit Filing:** The Cancer Letter. (2012, March 9). *Penn sues MSKCC President Thompson over rights to $261 million Agios*. Vol. 38 No. 10.(https://cdn.cancerhistoryproject.com/media/2012/03/10000000/TCL38-10.pdf)
- **Lawsuit Outcome:** Fierce Biotech. (2012, May 18). *Penn grabs Agios deal, settling high-profile IP theft claim*.

These sources are derived from a combination of academic profiles, oral history interviews, institutional archives, and professional news records. Below are the citations for the information provided across the scenes, quotes, human reality details, controversies, and distinctive findings, formatted in APA and separated by commas:

Advisory Board. (2018, October 4). *Were 3 scientists 'snubbed' by the Nobel Prize in medicine?* , Bluestone, J. A. (2016). Checkpoint blockade: The triumph of basics. *Science Signaling, 9*(415). , Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*.(https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf) , Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. , Cancer World. (n.d.). *The future of cancer treatment: Tak Mak, T-cell receptor, immunotherapies*. , El País. (2024, September 22). *Tak Mak, immunologist: 'The immune system is the orchestra for most of life's symphonies'*. , Fierce Biotech. (2012, May 18). *Penn grabs Agios deal, settling high-profile IP theft claim*. , Frontline Genomics. (2024). *Genome Giants: Tak Mak, Senior Scientist, Princess Margaret Cancer Centre*. , Government of Canada. (2002). *Tak W. Mak propels his University of Toronto lab to the forefront of cancer research*. Publications.gc.ca. , Immpress Magazine. (2024, June 3). *Ego, politics, and the scientific zeitgeist*. , Journal of Clinical Investigation. (2015, December 1). *Interview with James Allison*. , Mak, T. W. (2018). How I remember McCulloch. *Biographical Memoirs of Fellows of the Royal Society, 64*, 317–324. , National Academy of Sciences. (2011). Profile of Tak Wah Mak. *Proceedings of the National Academy of Sciences, 108*(27), 11169-11171. , OncoDaily. (2024). *40th Anniversary of the Discovery of the T Cell Receptor*. , ScienceDaily. (2019, February 8). *How the immune system 'thinks'*. , Signals Blog. (n.d.). *Right Turn: The awe-inspiring Tak Mak*. , The Cancer Letter. (2012, March 9). *Penn sues MSKCC President Thompson over rights to $261 million Agios*.(), The Princess Margaret Cancer Foundation. (2024, November 21). *Celebrating a revolution in immunology and cancer care*. , The Scientist. (2003, July 13). *The Holy Grail of Immunology*. , University Health Network. (2019, March 18). *Expecting the Unexpected*. .

Several dramatic details in the narrative of Tak Wah Mak's life are frequently repeated but

possess elements of narrative simplification, retrospective framing, or, in one case, mistaken identity.

Flagged “Myth-Risk” Details

Claim: Dr. Mak’s application to the University of Toronto was lost three times, leading to a late acceptance.

- **Where it appears:** Anecdotal summaries of Toronto-based researchers and common biographical queries.
- **Why sourcing is weak:** This specific event is documented as having happened to fellow Canadian genomics researcher Stephen Scherer, not Tak Mak. There is no mention of lost applications in Mak’s own extensive oral histories.
- **Best available evidence:** Stephen Scherer’s oral history from *Frontline Genomics*.

Claim: Tak Mak was formally designated a "national hero" by the Canadian public in 1987.

- **Where it appears:** Biographical profiles and news summaries reflecting on his decision to remain in Canada.
- **Why sourcing is weak:** The phrase "national hero for not leaving" originates as a self-referential observation by Mak himself, used in interviews to humorously characterize the local headlines of the era rather than an official designation or verifiable national consensus.
- **Best available evidence:** Mak's own interviews where he uses the phrase to illustrate the cultural differences between U.S. and Canadian science.

Claim: The cloning of the T-cell receptor (TCR) was an "intense race" with a singular winner.

- **Where it appears:** Scientific retrospectives and James Allison's Nobel lecture.
- **Why sourcing is weak:** While the search for the TCR was highly competitive, the primary discoverers (Mak and Mark Davis) were working on different species—human and mouse, respectively. Their results were published back-to-back in the same issue of *Nature*, suggesting a simultaneous and coordinated discovery rather than a "winner-takes-all" surprise.
- **Best available evidence:** The back-to-back articles in *Nature* published on March 8, 1984.

Claim: The 1984 hand-drawn sketch of the T-cell receptor was "perfectly correct" years before the structure was finalized.

- **Where it appears:** Profiles in *The Scientist* and laboratory lore.
- **Why sourcing is weak:** The sketch accurately predicted the basic heterodimeric form and intermolecular disulfide bonding, but the full 3D complexity of the TCR-CD3 complex—an assembly involving eight different protein subunits—was not resolved until decades later. The story tends to condense a multi-generational effort in

structural biology into a single "Eureka" drawing.

- **Best available evidence:** The physical sketch kept by Dr. Pamela Ohashi, which confirms Mak's accurate early intuition regarding the linkage of the α and β chains.

Claim: His grant to clone the TCR was rejected because it was considered "impossible."

- **Where it appears:** *PNAS* profiles and personal interviews.
- **Why sourcing is weak:** Reviewer comments are confidential, and the "impossible" label is Mak's characterization of the rejection rather than a verbatim transcript of the referee's report. The rejection was more likely based on the technical uncertainty of the "molecular subtraction" method at the time.

Best available evidence: Mak's consistent oral history across multiple decades confirms the rejection of the foundational grant.

SCENE #1: The Leap of Faith Summary: In a high-stakes effort to solve the "Holy Grail of Immunology," the laboratory uses an unconventional virology technique to clone the human T-cell receptor (TCR) beta chain, succeeding despite significant institutional barriers. Date/Place: March 8, 1984 (Publication); University of Toronto / Ontario Cancer Institute. People: Tak Wah Mak, Yusuke Yanagi (postdoctoral researcher), and Pamela Ohashi (student). Concrete details (quote-level if available): Facing a "rejected research grant" and working "against all odds," the team employed "molecular subtraction" to isolate T-cell-specific cDNA. Before the structure was fully confirmed, Mak took a "leap of faith" and hand-drew a sketch for Pamela Ohashi showing how he believed the α and β chains were held together by intermolecular disulfide bonding. Why it matters: It transformed immunology from observational biology into a molecular science and laid the groundwork for all modern T-cell therapies, including CAR-T and TCR-T. Evidence quality (primary/secondary; strength): Primary; Based on seminal Nature publications (1984) and personal accounts from lab members. Myth risk (low/medium/high + why): Medium; While the discovery was simultaneous with Mark Davis, the narrative of an "intense race" is sometimes heightened for drama, though they published back-to-back in the same issue.

SCENE #2: The Exploding Lymph Nodes Summary: The laboratory pioneers the use of "knockout mice" to determine the function of CTLA-4, discovering that rather than an activator, the molecule serves as a critical "brake" that prevents the immune system from destroying the body. Date/Place: 1995; Amgen Research Institute, Toronto. People: Tak Wah Mak, Paul Waterhouse (lab member), and Arlene Sharpe (independent collaborator/competitor). Concrete details (quote-level if available): The breakthrough came

when researchers observed the physical phenotype of mice lacking the CTLA-4 gene. The animals became hyper-inflamed and "turned into a big lymph node" because their T-cells "couldn't stop dividing." This provided "absolute genetic proof" of the molecule's inhibitory function. Why it matters: This discovery identified the first immune "checkpoint," providing the biological rationale for checkpoint inhibitor drugs like Ipilimumab, which revolutionized cancer treatment. Evidence quality (primary/secondary; strength): Primary/Secondary; Documented in Science (1995) and confirmed in later Nobel Prize-related retrospectives. Myth risk (low/medium/high + why): Low; The dramatic phenotype of the knockout mice is a well-documented turning point in functional genomics.

SCENE #3: The National Hero's Choice Summary: Following the international acclaim of the TCR discovery, Mak faces a pivotal career crossroad, ultimately choosing to remain in Canada despite lucrative offers to lead top-tier American institutions. Date/Place: 1987; Toronto / USA. People: Tak Wah Mak. Concrete details (quote-level if available): Mak was offered the positions of Head of Pathology at Yale University and Director of the Howard Hughes Medical Institute. He turned them down, leading to headlines calling him a "national hero for not leaving." He later remarked that science in the United States felt like a "contact sport" and preferred the innovative, less hyper-competitive environment of Toronto. Why it matters: This decision solidified Toronto as a global center for immunology and genetics, ensuring that his laboratory's "seeds"—over 150 trainees—would continue to develop within the Canadian research ecosystem. Evidence quality (primary/secondary; strength): Secondary; Based on biographical profiles and interviews reflecting on his career trajectory and philosophy. Myth risk (low/medium/high + why): Low; The job offers and Mak's subsequent "national hero" label are widely cited in his institutional biographies.

Sources:

Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*. https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf , Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. https://www.cdnmedhall.ca/laureates/takmak , Tak Wah Mak. (2024). In *Wikipedia*. https://en.wikipedia.org/wiki/Tak_Wah_Mak , The Scientist. (2003, July 13). *The Holy Grail of Immunology*. https://www.the-scientist.com/the-holy-grail-of-immunology-51335 , Allison, J. P. (2018). *Nobel Lecture: Immune Checkpoint Blockade in Cancer Therapy*. https://www.nobelprize.org/uploads/2018/10/allison-lecture.pdf , Oldfield, J. (2019, May 23). *'Discover everything there is': U of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research*. University of Toronto News. https://www.utoronto.ca/news/discover-everything-there-u-t-s-tak-mak-awarded-prestigious-gold-leaf-prize-pioneering-research , Government of Canada. (2002). *Tak W. Mak propels his University of Toronto lab to the forefront of cancer research*. Publications.gc.ca.

https://publications.gc.ca/collections/collection_2022/isde-ised/iu4/Iu4-8-2002-eng.pdf , National Academy of Sciences. (n.d.). *Profile of Tak Wah Mak*. Proceedings of the National Academy of Sciences. https://www.pnas.org/doi/10.1073/pnas.1116912108 , Cancer World. (n.d.). *The future of cancer treatment: Tak Mak, T-cell receptor, immunotherapies*. https://cancerworld.net/the-future-of-cancer-treatment-tak-mak-t-cell-receptor-immunotherapies/ , OncoDaily. (2024). *40th Anniversary of the Discovery of the T Cell Receptor*. https://oncodaily.com/insight/tcr179986 , University Health Network. (2023, March 14). *Honouring Extraordinary Achievement*. https://www.uhnresearch.ca/news/honouring-extraordinary-achievement

1. On the Centrality of Immunology

- **Exact Quote:** "All medical scientists have two lives. The second one begins when they realize that immunology is the orchestra for most of life's symphonies."
- **Context/Date:** 2023 interview with the Canadian Cancer Research Alliance (CCRA).
- **Where it appears:** CCRA Interview Transcript, *2023 Interview with Dr. Tak Wah Mak*.
- **Full Citation:** Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*. https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf

2. On Scientific Intuition (The "Leap of Faith")

- **Exact Quote:** "At the very beginning, when thoughts are still brewing in your mind, and the facts are not yet in, you have to take a leap of faith. And luckily, we turned out to be right."
- **Context/Date:** July 13, 2003; reflecting on his 1984 decision to sketch the predicted structure of the T-cell receptor before its final confirmation.
- **Where it appears:** *The Scientist* magazine, article titled "The Holy Grail of Immunology".
- **Full Citation:** The Scientist. (2003, July 13). *The Holy Grail of Immunology*. https://www.the-scientist.com/the-holy-grail-of-immunology-51335

3. On the Joy of Discovery

- **Exact Quote:** "When you're a kid you get to play with toys and figure things out, and you're not supposed to do that as an adult. But science is like toys for adults, and we get to discover everything there is."
- **Context/Date:** May 23, 2019; interview following the announcement of the CIHR Gold Leaf Prize for Discovery.
- **Where it appears:** University of Toronto News article, *'Discover everything there is': U*

of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research.

- **Full Citation:** Oldfield, J. (2019, May 23). *'Discover everything there is': U of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research*. University of Toronto News. https://www.utoronto.ca/news/discover-everything-there-u-t-s-tak-mak-awarded-prestigious-gold-leaf-prize-pioneering-research

4. On Retirement and Persistence

- **Exact Quote:** "I don't have anything left to prove - except that cancer is not cured yet."
- **Context/Date:** March 2023; statement regarding his continued research at age 76 after receiving the Pezcoller Foundation-AACR Award.
- **Where it appears:** Canadian Medical Hall of Fame Laureate Biography.
- **Full Citation:** Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. https://www.cdnmedhall.ca/laureates/takmak

5. On His Trainees (The "Seed Theory")

- **Exact Quote:** "The flowers of all the tomorrows are in the seeds of today. I like to think of my trainees as those seeds, who will set up their own labs and continue to spread flowers in the form of useful research discoveries."
- **Context/Date:** 2023; interview regarding his legacy and the training of over 150 postdoctoral fellows and students.
- **Where it appears:** CCRA Interview Transcript, *2023 Interview with Dr. Tak Wah Mak*.
- **Full Citation:** Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*. https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf

6. On Remaining in Canada

- **Exact Quote:** "Only in Canada does someone become a national hero for not leaving. Toronto is a good place to do science. It's less of a contact sport here than it is in the United States."
- **Context/Date:** 2002; commenting on his decision in 1987 to turn down prestigious leadership roles at Yale and the Howard Hughes Medical Institute.
- **Where it appears:** Government of Canada Publication, *Tak W. Mak Propels His University of Toronto Lab to the Forefront of Cancer Research*.
- **Full Citation:** Government of Canada. (2002). *Tak W. Mak propels his University of Toronto lab to the forefront of cancer research*. Publications.gc.ca. https://publications.gc.ca/collections/collection_2022/isde-ised/iu4/Iu4-8-2002-eng.p

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7. On Mentorship and Rigor

- **Exact Quote:** "McCulloch taught me to think differently, to have original ideas and integrate my thoughts. Till was a staunch, vigorous physicist, who insisted that all results be statistically significant and re-confirmed. The two together made an almost perfect mentorship for me."
- **Context/Date:** February 2019; reflecting on his early years at the Ontario Cancer Institute working with James Till and Ernest McCulloch.
- **Where it appears:** University of Toronto Faculty of Medicine News, *Tak Mak: Gold Leaf Prize and Life Science*.
- **Full Citation:** University of Toronto. (2019, February). *Tak Mak: Gold Leaf Prize and Life Science*. https://temertymedicine.utoronto.ca/news/tak-mak-gold-leaf-prize-and-life-science

As a scientist, Tak Wah Mak is characterized by a blend of Jesuit-influenced rigor, merchant-class pragmatism, and a deliberate avoidance of "scientific crowds."

Human Reality Details

- **Upbringing/Constraint:** Born into a family of silk merchants in post-war Hong Kong, Mak faced a culture where high-achieving youth were strictly expected to become doctors, lawyers, or businessmen; he chose biochemistry primarily to "avoid medical school".
- **Formative Experience/Jesuit Education:** His education by Jesuit priests at Wah Yan College instilled a "creed" of benevolence and fairness that shaped his eventual scientific principles of inquiry and persistence.
- **Obstacle/Early Struggle:** When he first arrived in the United States, his family had very little money, forcing him to cover his tuition by working as a gardener and a glassware washer in research facilities.
- **Working Style/Non-Conformity:** As a student, Mak displayed a pragmatic defiance of academic busywork, once solving only three out of 100 assigned math problems and arguing that solving the remaining 97 was unnecessary if his initial logic was correct.
- **Mentor Influence:** Mak's scientific rigor was forged by a "perfect" dual mentorship: Ernest McCulloch taught him to think intuitively and "big picture," while James Till, a physicist, insisted on absolute re-confirmation of every result.
- **Working Habit/Retreats:** He fostered a collaborative culture by hosting lab "leukaemia meetings" at family cottage retreats on Ahmic Lake, combining sailing and outdoor activities with intense biological discussion.

- **Scientific Philosophy/Solitude:** Mak intentionally avoids "scientific crowds," believing that a researcher must be "lonely" enough to retreat from the group and evaluate ideas with their own unique bias to find new paths.
- **Mentoring Style/Legacy:** He views his trainees as "seeds" rather than just employees, taking more pride in the 150+ researchers he has mentored into leadership roles than in his own individual awards.
- **Working Style/Intuition:** He is a proponent of taking a "leap of faith" in research, famously hand-sketching the T-cell receptor's alpha-beta chain structure for a student years before the physical evidence was finalized.
- **Cultural Identity/National Decision:** Mak rejected leadership offers from Yale and the Howard Hughes Medical Institute because he preferred Canada's less hyper-competitive environment, which he described as less of a "contact sport" than U.S. science.
- **Quirks/Proverbs:** He frequently frames complex scientific strategies using proverbs, such as the African proverb "If you want to go far, go together," to emphasize teamwork in tackling cancer.
- **Habit/Persistence:** At 79 years old, he continues to lead active research projects, famously stating he has nothing left to prove except that "cancer is not cured yet".

Tak Wah Mak's research provides a direct bridge to introductory biochemistry, specifically within the study of **Metabolism and Enzyme Catalysis**.

BIOL 3030 Bridge

- **Best-Fitting Topic:** Metabolism and Enzyme Catalysis (specifically the Citric Acid Cycle and Redox Cofactors).
- **Precise Contribution:** Dr. Mak discovered that mutations in the metabolic enzymes Isocitrate Dehydrogenase 1 and 2 (IDH1/2) confer a "neomorphic" gain-of-function activity, producing the oncometabolite D-2-hydroxyglutarate (D-2-HG) which drives cancer progression.
- **Teachable Bridge:** In introductory biochemistry, students learn that IDH is a key enzyme in cellular respiration that catalyzes the oxidative decarboxylation of isocitrate to α-ketoglutarate (α-KG) while producing $NADPH$. Mak's work demonstrates how a single amino acid substitution in the enzyme's active site causes it to consume $NADPH$ to reduce α-KG into D-2-HG. This rare metabolite then acts as a competitive inhibitor for other α-KG-dependent enzymes, such as DNA demethylases, illustrating how a specific change in enzyme kinetics and substrate specificity can rewire a cell's entire epigenetic landscape to promote malignancy.

Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*.(https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf) , Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. https://www.cdnmedhall.ca/laureates/takmak , El País. (2024, September 22). *Tak Mak,*

immunologist: 'The immune system is the orchestra for most of life's symphonies'. https://english.elpais.com/health/2024-09-22/tak-mak-immunologist-the-immune-system-is-the-orchestra-for-most-of-lifes-symphonies.html , Frontline Genomics. (2024). *Genome Giants: Tak Mak, Senior Scientist, Princess Margaret Cancer Centre*. https://frontlinegenomics.com/genome-giants-tak-mak-senior-scientist-princess-margaret-cancer-centre/ , Government of Canada. (2002). *Tak W. Mak propels his University of Toronto lab to the forefront of cancer research*. Publications.gc.ca. https://publications.gc.ca/collections/collection_2022/isde-ised/iu4/Iu4-8-2002-eng.pdf , Mak, T. W. (2018). How I remember McCulloch. *Biographical Memoirs of Fellows of the Royal Society, 64*, 317–324. https://doi.org/10.1098/rsbm.2017.0033 , National Academy of Sciences. (2011). *Profile of Tak Wah Mak*. *Proceedings of the National Academy of Sciences, 108*(2), 11169-11171. https://doi.org/10.1073/pnas.1116912108 , Oldfield, J. (2019, May 23). *'Discover everything there is': U of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research*. University of Toronto News. https://www.utoronto.ca/news/discover-everything-there-u-t-s-tak-mak-awarded-prestigious-gold-leaf-prize-pioneering-research , Spanish National Cancer Research Center (CNIO). (2023, March 21). *Tak W. Mak, immunologist: "I am not doing science for recognition"*. https://www.cnio.es/en/news/tak-w-mak-immunologist-i-am-not-doing-science-for-recognition/ , The Scientist. (2003, July 13). *The Holy Grail of Immunology*. https://www.the-scientist.com/the-holy-grail-of-immunology-51335 , University of Toronto. (2019, February). *Tak Mak: Gold Leaf Prize and Life Science*. https://temertymedicine.utoronto.ca/news/tak-mak-gold-leaf-prize-and-life-science , Wikipedia. (2024). *Tak Wah Mak*.(https://en.wikipedia.org/wiki/Tak_Wah_Mak) .

Scientist's Analogy

When describing the metabolic mutations in cancer cells that allow them to adapt to their environment and evade death, Mak characterizes this strategic biological shift as "**shaking hands with the devil**." He also frequently employs a mechanical analogy for the immune system, where the T-cell receptor is the **ignition**, the CD28 molecule is the **accelerator**, and the CTLA-4 molecule (which he co-discovered) is the **brake**.

Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*. https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf , Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. https://www.cdnmedhall.ca/laureates/takmak , Wikipedia. (2024). *Tak Wah Mak*. https://en.wikipedia.org/wiki/Tak_Wah_Mak , The Scientist. (2003, July 13). *The Holy Grail of Immunology*. https://www.the-scientist.com/the-holy-grail-of-immunology-51335 , Allison, J. P. (2018). *Nobel Lecture: Immune Checkpoint Blockade in Cancer Therapy*. https://www.nobelprize.org/uploads/2018/10/allison-lecture.pdf , Oldfield, J. (2019, May 23). *'Discover everything there is': U of T's Tak Mak awarded prestigious Gold Leaf Prize for pioneering research*. University of Toronto News. https://www.utoronto.ca/news/discover-everything-there-u-t-s-tak-mak-awarded-prestigious-gold-leaf-prize-pioneering-research , Government of Canada. (2002). *Tak W. Mak propels*

his University of Toronto lab to the forefront of cancer research. Publications.gc.ca. https://publications.gc.ca/collections/collection_2022/isde-ised/iu4/Iu4-8-2002-eng.pdf , National Academy of Sciences. (n.d.). *Profile of Tak Wah Mak*. Proceedings of the National Academy of Sciences. https://www.pnas.org/doi/10.1073/pnas.1116912108 , Cancer World. (n.d.). *The future of cancer treatment: Tak Mak, T-cell receptor, immunotherapies*. https://cancerworld.net/the-future-of-cancer-treatment-tak-mak-t-cell-receptor-immunotherapies/ , OncoDaily. (2024). *40th Anniversary of the Discovery of the T Cell Receptor*. https://oncodaily.com/insight/tcr179986 , University Health Network. (2023, March 14). *Honouring Extraordinary Achievement*. https://www.uhnresearch.ca/news/honouring-extraordinary-achievement , The Princess Margaret Cancer Foundation. (2024, November 21). *Celebrating a revolution in immunology and cancer care*. https://thepmcf.ca/media-supporter-stories/latest-news/2024/celebrating-a-revolution-in-immunology-and-cancer-care/ , Chinese Australia. (2024). *Shunde silk production*. https://chineseaustralia.org/2024/01/.

Tak Wah Mak's career has been marked by several significant priority disputes, institutional barriers, and documented investigations that reflect the high-stakes nature of modern immunology and oncology.

1. The T-Cell Receptor Priority Race and Funding Rejection

- **What's agreed upon:** In the early 1980s, the race to clone the T-cell receptor (TCR) was considered the "Holy Grail of Immunology." Most established groups, including those led by Ellis Reinherz and James Allison, attempted to isolate the receptor via protein purification. Tak Mak and Mark Davis ultimately won the race by using the "molecular subtraction" technique at the RNA/DNA level.
- **What's disputed:** Mak faced significant institutional skepticism regarding his approach. His initial grant proposal to fund the search was flatly rejected, as reviewers considered the proposed molecular subtraction method "impossible" or unlikely to succeed. Furthermore, while Mak and Davis are credited with the genetic discovery, other scientists like Reinherz had previously identified the protein's biochemical components, leading to long-standing discussions regarding who provided the "true" discovery.
- **Best sources:** *Nature* (1984) for the discovery; *PNAS* for the rejected grant narrative.

2. The CTLA-4 Function Controversy and the 2018 Nobel "Snub"

- **What's agreed upon:** James Allison and Tasuku Honjo received the 2018 Nobel Prize for discovering cancer therapy via immune checkpoint blockade. Mak's lab, alongside Arlene Sharpe's, provided the "absolute genetic proof" in 1995 that CTLA-4 acts as an

inhibitory "brake" on T-cells.

- **What's disputed:** A priority dispute existed in the mid-1990s regarding whether CTLA-4 was an activator or an inhibitor. Earlier researchers, such as Peter Linsley, initially suggested CTLA-4 was a co-stimulatory "gas pedal" similar to CD28. Mak and Sharpe independently overturned this "heretical" view. When the 2018 Nobel was awarded to Allison (who applied the brake theory to cancer), many experts were "shocked" that the genetic discoverers—specifically Mak, Sharpe, and Gordon Freeman—were excluded due to the Nobel "Rule of 3."
- **Best sources:** *Science* (1995) for the knockout data; *STAT News* and *New York Times* for the 2018 Nobel exclusion controversy.

3. Investigation of Alleged Scientific Misconduct

- **What's agreed upon:** Tak Mak and his high-profile protégé, Josef Penninger, were the subjects of a documented institutional investigation.
- **What's disputed:** The investigation focused on alleged scientific misconduct. While Penninger eventually moved to lead the Institute of Molecular Biotechnology in Austria, the publicly documented fallout was described by peers as a significant "falling-out" between elite scientists that captured the attention of the Canadian research community.
- **Best sources:** *Immpress Magazine*; *Schneider Shorts* (2025 Retrospective).

4. Agios Intellectual Property Lawsuit

- **What's agreed upon:** Tak Mak co-founded Agios Pharmaceuticals with Craig Thompson and Lewis Cantley to target cancer metabolism. In 2011-2012, the University of Pennsylvania and the Abramson Institute filed lawsuits against Thompson and the company.
- **What's disputed:** The lawsuit alleged that Thompson had "absconded" with intellectual property developed at Penn related to IDH1 and IDH2 mutations to found Agios. While Thompson was the primary target of the "unscrupulous" label in the filings, the dispute threatened the reputation and financial foundations of the company Mak helped build. The defense argued that authorship of papers did not equate to patentable "invention" under the specific circumstances.
- **Best sources:** *The Cancer Letter* archive.

Here are the APA-formatted sources for the complications, disputes, and controversies involving Tak Wah Mak:

1. The T-Cell Receptor Priority Race and Funding Rejection

- **Grant Rejection:** National Academy of Sciences. (2011). Profile of Tak Wah Mak. *Proceedings of the National Academy of Sciences, 108*(27), 11169-11171.

- **Discovery Priority:** Yanagi, Y., Yoshikai, Y., Leggett, K., Clark, S. P., Aleksander, I., & Mak, T. W. (1984). A human T cell-specific cDNA clone encodes a protein having extensive homology to immunoglobulin chains. *Nature, 308*(5955), 145-149.
- **Historical Competition:** Odaka, C. (2020). PD-1: Its discovery, involvement in cancer immunotherapy, and current research. *Cells, 9*(6), 1376.

2. The CTLA-4 Function Controversy and the 2018 Nobel "Snub"

- **Genetic Proof:** Waterhouse, P., Penninger, J. M., Holliday, E., Ohashi, P. S., & Mak, T. W. (1995). Lymphoproliferative disorders with early lethality in mice deficient in CTLA-4. *Science, 270*(5238), 985-988.
- **Nobel Exclusion Controversy:** Advisory Board. (2018, October 4). *Were 3 scientists 'snubbed' by the Nobel Prize in medicine?*
- **Scientific Background of Snub:** Bluestone, J. A. (2016). Checkpoint blockade: The triumph of basics. *Science Signaling, 9*(415).

3. Investigation of Alleged Scientific Misconduct

- **Investigation Details:** Immpress Magazine. (2024, June 3). *Ego, politics, and the scientific zeitgeist*.
- **Legacy Context:** Schneider, L. (2025, November 7). Schneider Shorts 7.11.2025: No data were falsified or fabricated in any way. *For Better Science*.

4. Agios Intellectual Property Lawsuit

- **Lawsuit Filing:** The Cancer Letter. (2012, March 9). *Penn sues MSKCC President Thompson over rights to $261 million Agios*. Vol. 38 No. 10.(https://cdn.cancerhistoryproject.com/media/2012/03/10000000/TCL38-10.pdf)
- **Lawsuit Outcome:** Fierce Biotech. (2012, May 18). *Penn grabs Agios deal, settling high-profile IP theft claim*.

These sources are derived from a combination of academic profiles, oral history interviews, institutional archives, and professional news records. Below are the citations for the information provided across the scenes, quotes, human reality details, controversies, and distinctive findings, formatted in APA and separated by commas:

Advisory Board. (2018, October 4). *Were 3 scientists 'snubbed' by the Nobel Prize in medicine?* , Bluestone, J. A. (2016). Checkpoint blockade: The triumph of basics. *Science Signaling, 9*(415). , Canadian Cancer Research Alliance. (2023). *2023 Interview with Dr. Tak Wah Mak*.(https://www.ccra-acrc.ca/wp-content/uploads/2024/02/2023_Interview_TWMak_rev.pdf) , Canadian Medical Hall of Fame. (n.d.). *Tak Wah Mak*. , Cancer World. (n.d.). *The future of cancer treatment: Tak Mak, T-cell receptor, immunotherapies*. , El País. (2024, September 22).

Tak Mak, immunologist: 'The immune system is the orchestra for most of life's symphonies'. , Fierce Biotech. (2012, May 18). *Penn grabs Agios deal, settling high-profile IP theft claim*. , Frontline Genomics. (2024). *Genome Giants: Tak Mak, Senior Scientist, Princess Margaret Cancer Centre*. , Government of Canada. (2002). *Tak W. Mak propels his University of Toronto lab to the forefront of cancer research*. Publications.gc.ca. , Immpress Magazine. (2024, June 3). *Ego, politics, and the scientific zeitgeist*. , Journal of Clinical Investigation. (2015, December 1). *Interview with James Allison*. , Mak, T. W. (2018). How I remember McCulloch. *Biographical Memoirs of Fellows of the Royal Society, 64*, 317–324. , National Academy of Sciences. (2011). Profile of Tak Wah Mak. *Proceedings of the National Academy of Sciences, 108*(27), 11169-11171. , OncoDaily. (2024). *40th Anniversary of the Discovery of the T Cell Receptor*. , ScienceDaily. (2019, February 8). *How the immune system 'thinks'*. , Signals Blog. (n.d.). *Right Turn: The awe-inspiring Tak Mak*. , The Cancer Letter. (2012, March 9). *Penn sues MSKCC President Thompson over rights to $261 million Agios*.(), The Princess Margaret Cancer Foundation. (2024, November 21). *Celebrating a revolution in immunology and cancer care*. , The Scientist. (2003, July 13). *The Holy Grail of Immunology*. , University Health Network. (2019, March 18). *Expecting the Unexpected*. .

Several dramatic details in the narrative of Tak Wah Mak's life are frequently repeated but possess elements of narrative simplification, retrospective framing, or, in one case, mistaken identity.

Flagged "Myth-Risk" Details

Claim: Dr. Mak's application to the University of Toronto was lost three times, leading to a late acceptance.

- **Where it appears:** Anecdotal summaries of Toronto-based researchers and common biographical queries.
- **Why sourcing is weak:** This specific event is documented as having happened to fellow Canadian genomics researcher Stephen Scherer, not Tak Mak. There is no mention of lost applications in Mak's own extensive oral histories.
- **Best available evidence:** Stephen Scherer's oral history from *Frontline Genomics*.

Claim: Tak Mak was formally designated a "national hero" by the Canadian public in 1987.

- **Where it appears:** Biographical profiles and news summaries reflecting on his decision to remain in Canada.
- **Why sourcing is weak:** The phrase "national hero for not leaving" originates as a self-referential observation by Mak himself, used in interviews to humorously characterize the local headlines of the era rather than an official designation or verifiable national consensus.
- **Best available evidence:** Mak's own interviews where he uses the phrase to illustrate

the cultural differences between U.S. and Canadian science.

Claim: The cloning of the T-cell receptor (TCR) was an "intense race" with a singular winner.

- **Where it appears:** Scientific retrospectives and James Allison's Nobel lecture.
- **Why sourcing is weak:** While the search for the TCR was highly competitive, the primary discoverers (Mak and Mark Davis) were working on different species—human and mouse, respectively. Their results were published back-to-back in the same issue of *Nature*, suggesting a simultaneous and coordinated discovery rather than a "winner-takes-all" surprise.
- **Best available evidence:** The back-to-back articles in *Nature* published on March 8, 1984.

Claim: The 1984 hand-drawn sketch of the T-cell receptor was "perfectly correct" years before the structure was finalized.

- **Where it appears:** Profiles in *The Scientist* and laboratory lore.
- **Why sourcing is weak:** The sketch accurately predicted the basic heterodimeric form and intermolecular disulfide bonding, but the full 3D complexity of the TCR-CD3 complex—an assembly involving eight different protein subunits—was not resolved until decades later. The story tends to condense a multi-generational effort in structural biology into a single "Eureka" drawing.
- **Best available evidence:** The physical sketch kept by Dr. Pamela Ohashi, which confirms Mak's accurate early intuition regarding the linkage of the α and β chains.

Claim: His grant to clone the TCR was rejected because it was considered "impossible."

- **Where it appears:** *PNAS* profiles and personal interviews.
- **Why sourcing is weak:** Reviewer comments are confidential, and the "impossible" label is Mak’s characterization of the rejection rather than a verbatim transcript of the referee's report. The rejection was more likely based on the technical uncertainty of the "molecular subtraction" method at the time.
- **Best available evidence:** Mak’s consistent oral history across multiple decades confirms the rejection of the foundational grant.